



Department of
Electronic
Systems
Engineering



Department of Electronic
Systems Engineering

Indian Institute of Science
Bangalore 560 012 INDIA

Tel : +91 80 2293 3086

Email: haresh@iisc.ac.in

Date : 15.05.2026

To Whom It May Concern,

This is to certify that **Kumavath Sai Charitha** completed a 4-month internship at the Department of Electronic Systems Engineering (DESE), IISc Bangalore, from January 12, 2026, to May 15, 2026.

Under the supervision of Prof. Haresh Dagale, Charitha contributed to the **FirmwareHive** research project, focusing on **Embedded Systems and Cyber Security**. Her technical contributions included:

- **Security Frameworks:** Implemented the FirmwareHive Tree-of-Agents (ToA) framework, utilizing Recursive Delegation and parallel agent execution for autonomous analysis.
- **Vulnerability Assessment:** Analyzed real-world firmware (TP-Link, GE_B30, REx670), generating over 159 findings with risk scores reaching 9.0.
- **Data Tracing:** Performed multi-file tracing for credential discovery, CVE detection, and taint-style vulnerability analysis.
- **Local LLM Integration:** Integrated the Qwen model via Ollama for privacy-preserving analysis of buffer overflows and command injections.
- **Technical Proficiency:** Utilized Ghidra and Radare2 for deep binary analysis and dynamic task delegation.

Charitha is a disciplined and highly capable researcher. Her ability to navigate complex codebases makes her an excellent candidate for any role in cybersecurity or systems engineering.

HARESH DAGALE
Principal Research Scientist
Dept. of Electronic Systems Engineering
Indian Institute of Science
Bangalore - 560 012

Yours sincerely,
Mr. Haresh Dagale,
Principal Research Scientist.